



Towards Predictive Science for Controllable Learning Systems

Previous Work, Curiosity and Future Vision

Keyu Wang
September, 2026

Background

➤ Education

M.Sc. of Machine Learning
University of Tuebingen

Oct. 2024 – May 2027 (Expected)
Tuebingen, Germany

B.Eng of Artificial Intelligence
Southeast University

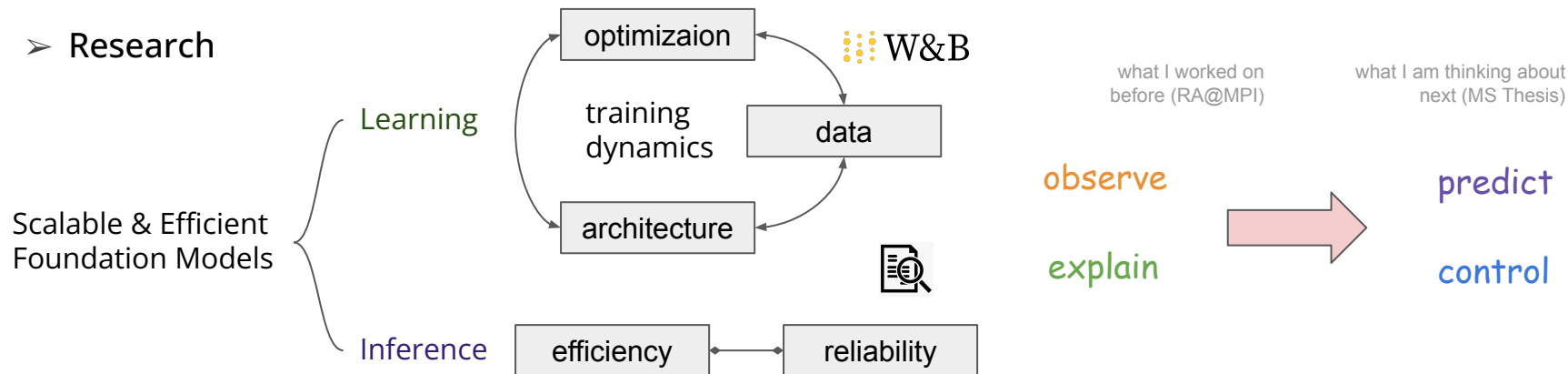
Sep. 2020 – Jun. 2024
Nanjing, China

➤ Experience

Research Assistant
Max Planck Institute for Intelligent Systems

Dec. 2025 – Jan. 2027 (Expected)
Tuebingen, Germany

➤ Research





MAX PLANCK INSTITUTE
FOR INTELLIGENT SYSTEMS



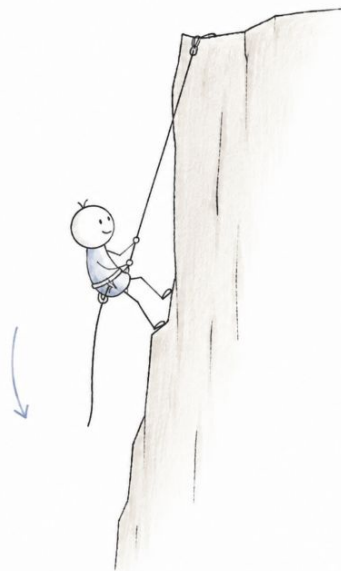
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Towards Effective Depth Scaling in LLMs

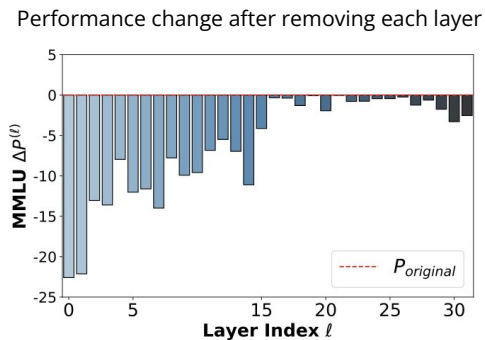
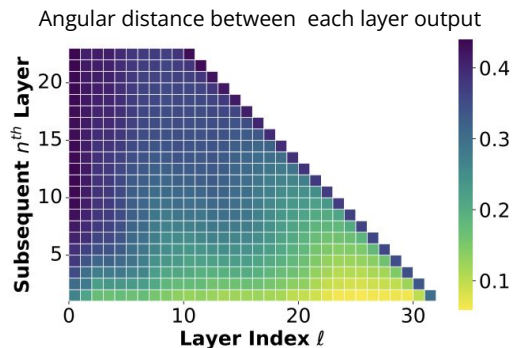
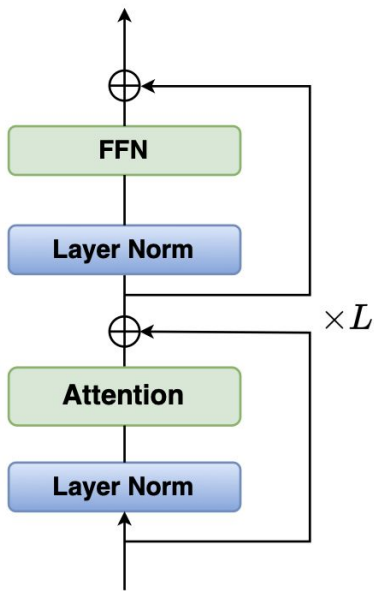
My Recent Research Line

Keyu Wang
Advisor: Dr. Shiwei Liu

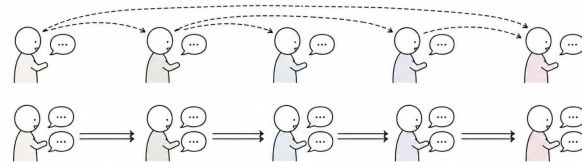
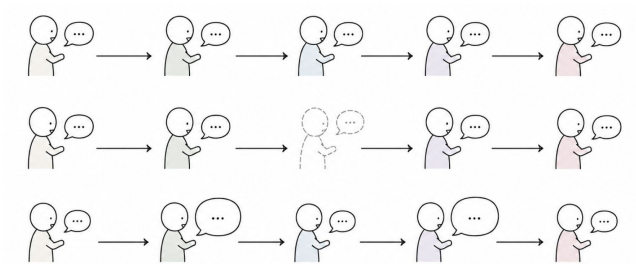
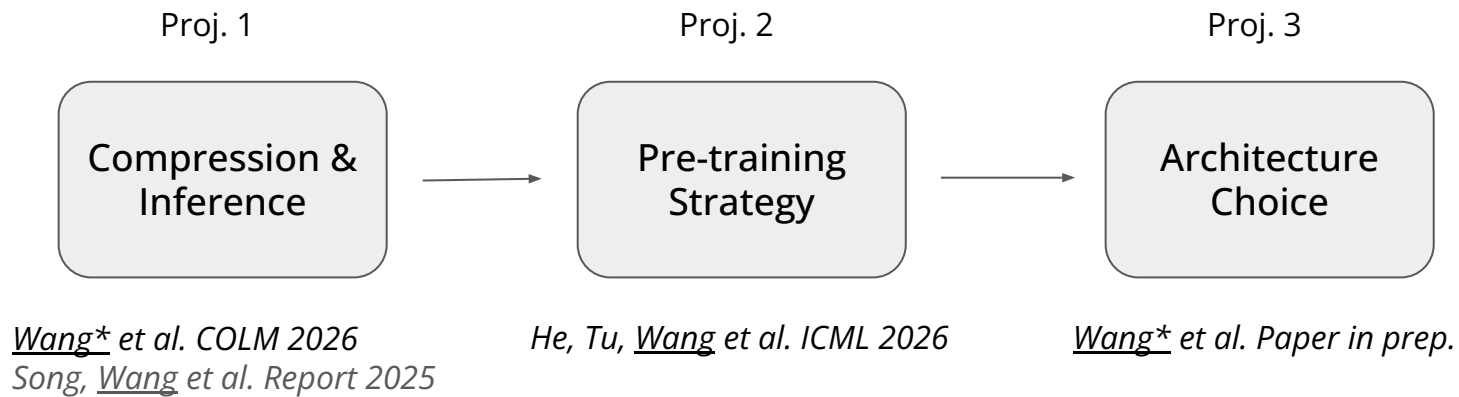


Motivation: depth scaling pays diminishing returns

- Pre-LN dominates in modern LLMs
— simple yet **stable** at scale
- But, Pre-LN faces the **Curse of Depth**
— deep layers become similar and ineffective



Outline



Project 1: Conference on Language Modeling - 2026

When Fewer Layers Break More Chains: Layer Pruning Harms Test-Time Scaling in LLMs

Keyu Wang^{1,2,3*} **Tian Lyu**^{4*} **Guinan Su**^{1,2,3} **Lu Yin**⁵ **Marco Canini**⁴
Jonas Geiping^{1,2,3} **Shiwei Liu**^{1,2,3}

¹Max Planck Institute for Intelligent Systems ²ELLIS Institute Tübingen

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⁵University of Surrey

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Project 1: Motivation

- The **curse** of depth seems to be a **blessing** for layer pruning...

ShortGPT: Layers in Large Language Models are **More Redundant** Than You Expect

Xin Men*
Baichuan Inc.

Mingyu Xu*
Baichuan Inc.

Qingyu Zhang*
ISCAS

Bingning Wang†
Baichuan Inc.

Hongyu Lin
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Yaojie Lu
ISCAS

Xianpei Han
ISCAS

Weipeng Chen
Baichuan Inc.

PRUNE&COMP: **Free Lunch** for Layer-Pruned LLMs via Iterative Pruning with Magnitude Compensation

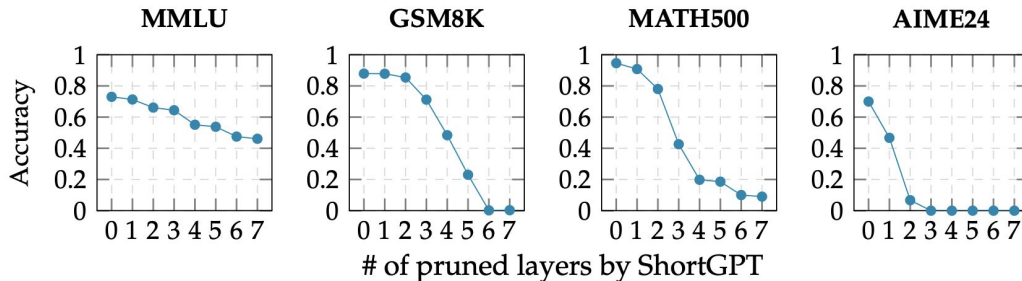
Xinrui Chen¹, Hongxing Zhang², Fanyi Zeng¹, Yongxian Wei¹, Yizhi Wang¹,
Xitong Ling¹, Guanghao Li¹, Chun Yuan^{1*}

¹ Shenzhen International Graduate School, Tsinghua University

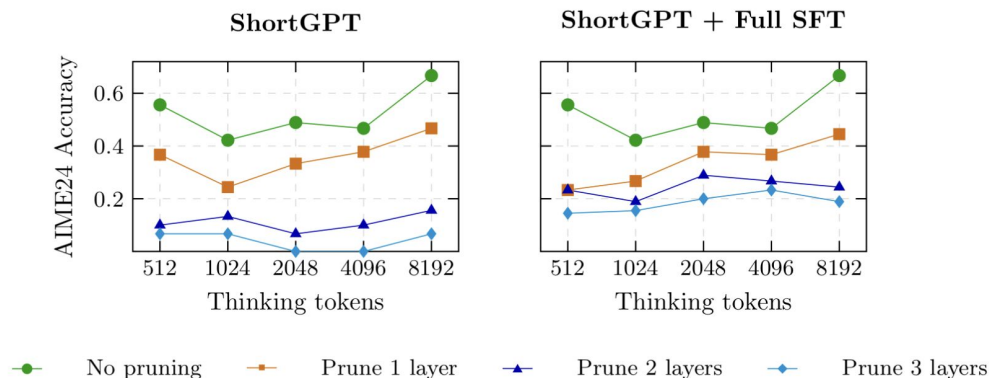
² School of Information Science and Technology, Guangdong University of Foreign Studies
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Recent studies reveal that the deeper layers (Transformer blocks) in modern LLMs tend to be less effective than the earlier ones (Gromov et al., 2024; Li et al., 2024b; Men et al., 2024; Yin et al., 2024). On the one hand, this interesting observation provides an effective indicator for LLM compression. For instance, we can compress deeper layers significantly more (Dumitru et al., 2024; Lu et al., 2024; Yin et al., 2024) to achieve high compression ratios. Even more aggressively, entire deep layers can be pruned completely without compromising performance (Muralidharan et al., 2024; Siddiqui et al., 2024).

- But layer pruning **collapses** complex reasoning!
Were we deceived by previous benchmarks of multiple-choice questions?



Project 1: Key Findings



- Sequential test-time scaling is highly **fragile** under layer pruning
- SFT can't lasso the runaway stallion

[AIME24]

Let p be the least prime number for which there exists a positive integer n such that $n^4 + 1$ is divisible by p^2 . Find the least positive integer m such that $m^4 + 1$ is divisible by p^2 . **Answer:** 110

Before Finetuning (Pruned Only)

The smallest prime p congruent to 1 modulo 8 is 17... 17 divides 83521-17=83504. Therefore, 17 divides 83521-17, so 17 divides 83521-17. Therefore, 17 divides 83521-17. Therefore, 17 divides 83521-17...

After Finetuning (Pruned + SFT)

Let check small primes. If $p = 2$, $n^4 + 1 \equiv 2 \pmod{4}$. If $p = 3$, $n^4 + 1 \equiv 2 \pmod{9}$ If $p = 11$, $n^4 \equiv -1 \equiv 10 \pmod{11}$, if $n = 5$, $n^4 + 1 = 626 \equiv 10 \pmod{11}$... The least prime p for which there exists n such that $p^2 \mid n^4 + 1$ is $p = 11$. When $p = 11$, we need $n^4 \equiv 10 \pmod{11}$. If $n = 5$, $n^4 = 625 \equiv 10 \pmod{11}$... Now we need $m^4 + 1 \equiv 0 \pmod{121}$. If $m = 5$, $5^4 + 1 = 626 \equiv 120 \pmod{121}$. Final Answer: 5.

Project 2: International Conference on Machine Learning - 2026

One LR Doesn't Fit All: Heavy-Tail Guided Layerwise Learning Rates for LLMs

Di He^{1,2,3} **Songjun Tu**^{2,3} **Keyu Wang**⁴ **Lu Yin**^{5†} **Shiwei Liu**^{6,7,8†}

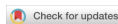
Project 2: Motivation

- Transformer layers are naturally trained **heterogeneously**

ARTICLE

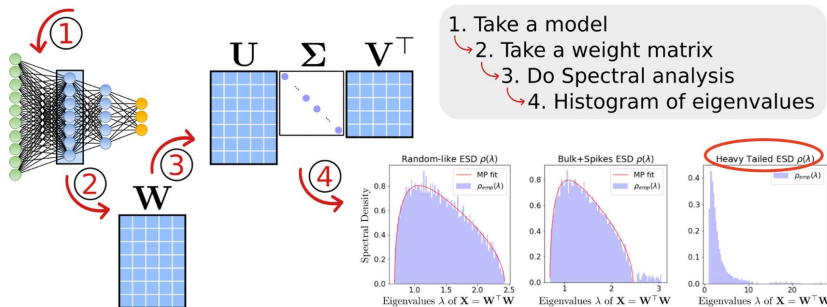
<https://doi.org/10.1038/s41467-021-24025-8>

OPEN



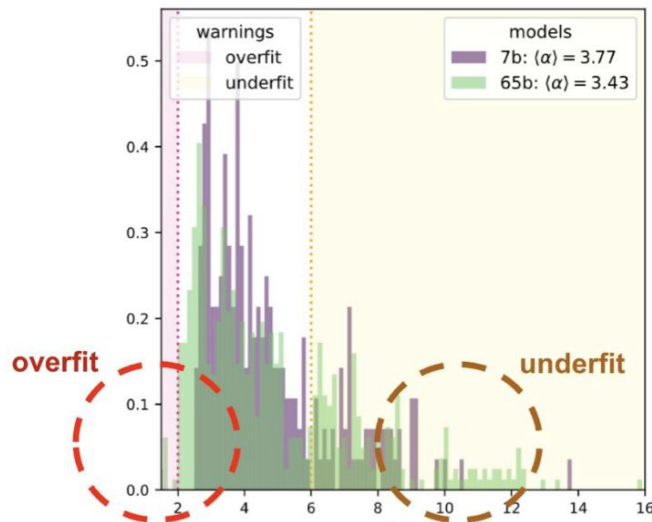
Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data

Charles H. Martin¹, Tongsu (Serena) Peng¹ & Michael W. Mahoney²



$$\rho_{\text{emp}}(\lambda) \sim \lambda^{-\alpha}$$

Llama



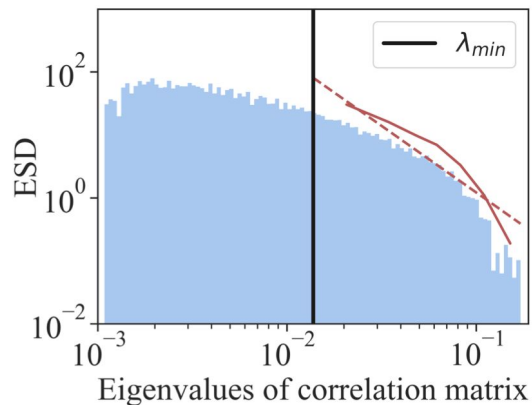
Project 2: Algorithm & Key Results

➤ Measure **Heavy-tail** structure per-layer

$$p(\lambda) \propto \lambda^{-\alpha}$$

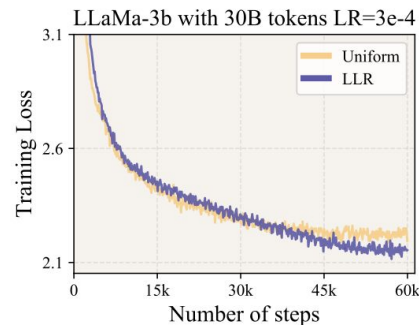
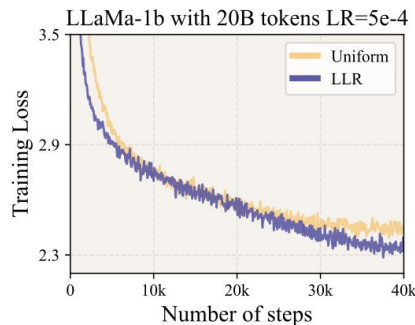
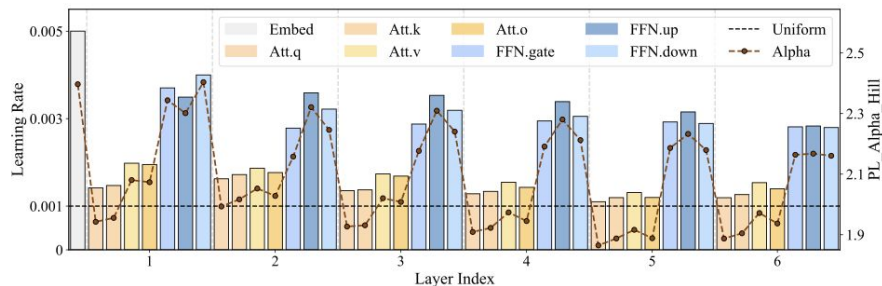
↑
estimate

$$\text{PL_Alpha_Hill} = 1 + \frac{k}{\sum_{i=1}^k \ln \frac{\lambda_{n-i+1}}{\lambda_{n-k}}}$$



➤ Transfer to **layerwise LR**

$$\eta_i^T = \eta \left[1 + (s-1) \frac{\alpha_i^T - \alpha_{min}^T}{\alpha_{max}^T - \alpha_{min}^T} \right]$$



Project 3: Paper is in cooking

Architecture-dependent Aspect Ratio Scaling

Contributors 4



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huangyangyi Yangyi Huang



davidgonmar David González-Ma...



Joluck J.L



Shiwei Liu

Faculty@ELLIS Institute Tübingen & MPI for Intelligent Systems
在 tue.ellis.eu 的电子邮件经过验证 - [首页](#)

[Deep Learning](#) [AI Modeling](#) [Low-dimensional Learning](#) [LLMs](#)



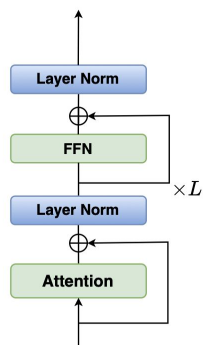
Weiyang Liu

CUHK
Verified email at cse.cuhk.edu.hk - [Homepage](#)

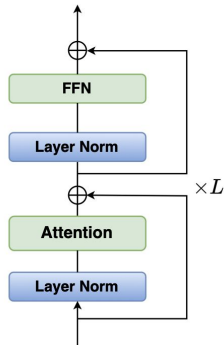
[Machine Learning](#)

Project 3: Motivation

- Which one scales its depth better?

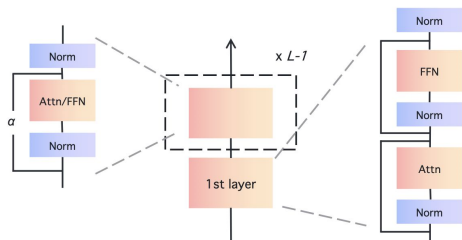


Post-LN

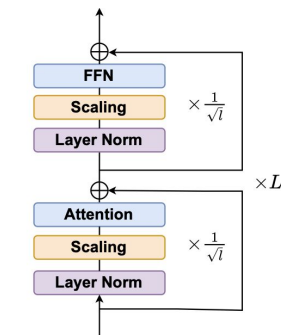


Pre-LN

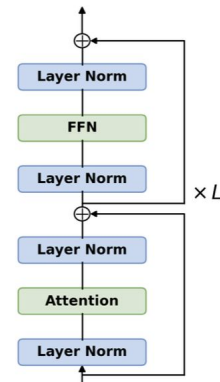
Dominate in modern LLMs



KEEL

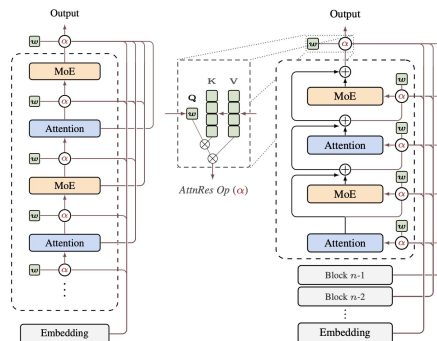


LayerNorm Scaling (LNS)



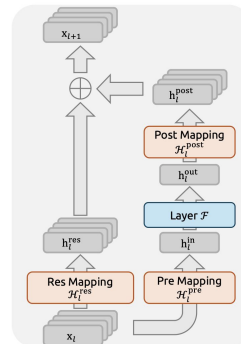
Sandwich-LN

Applied in Gemma 3/4



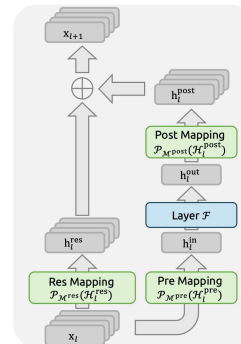
Attention Residuals (AttnRes, left: Full, right: Block)

Applied in Kimi K3



Hyper Connections (HC)

Applied (Modified) in Owen 3.8

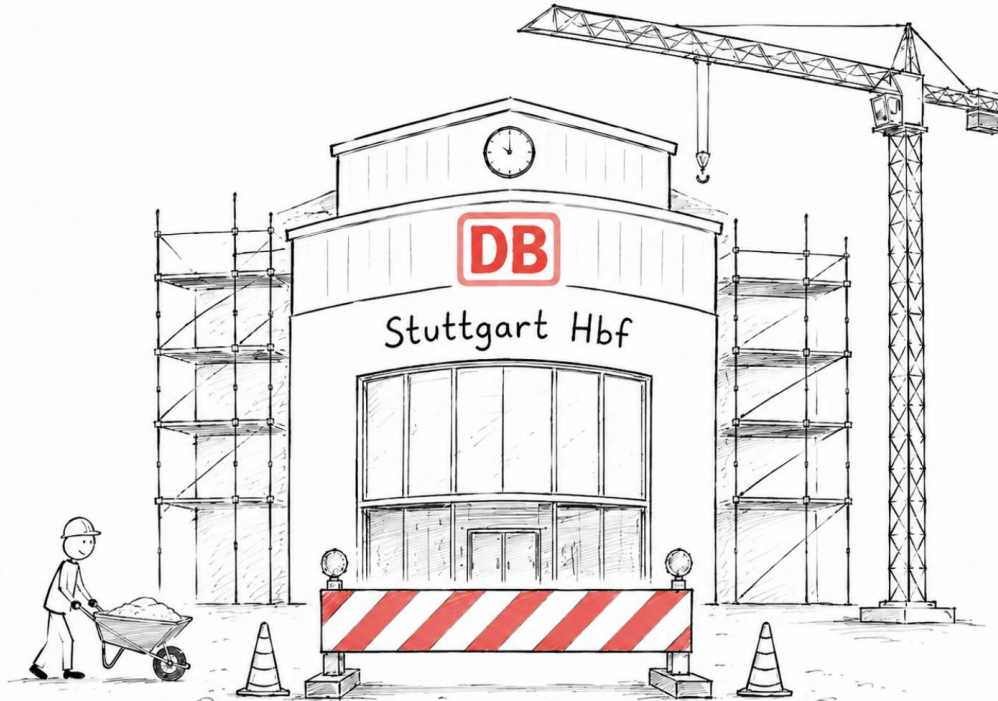


manifold-constrained HC (mHC)

Applied in Deepseek V4

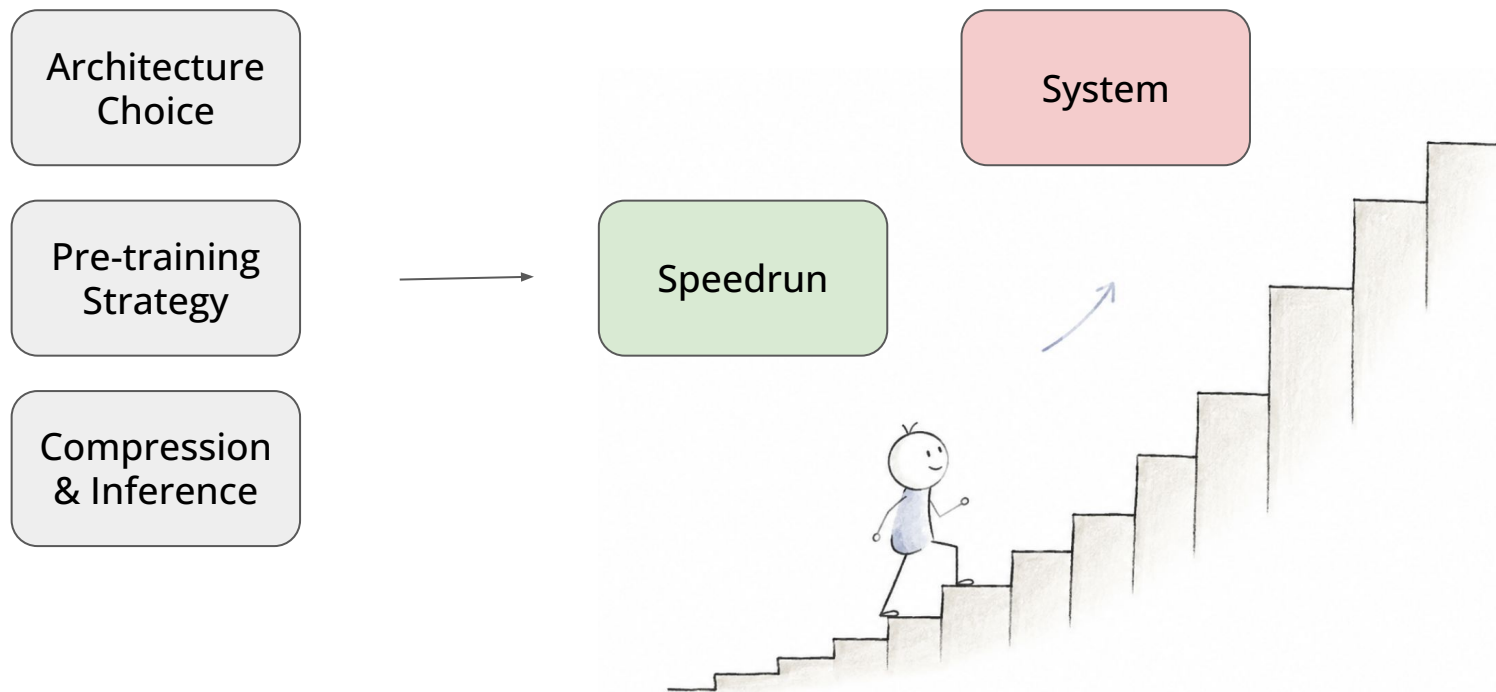


Project 3: Paper is coming soon!



Towards Effective Depth Scaling in LLMs

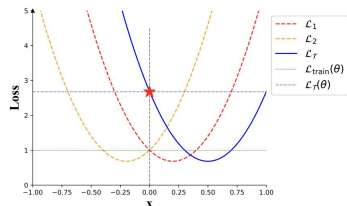
2025.06 - 2026.12



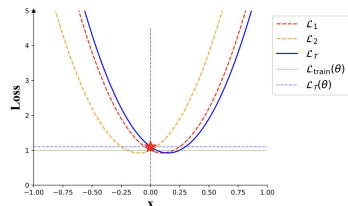
Future Vision

BTW I'm exploring my master's thesis direction about model dynamics & continual learning.
Happy to talk.

➤ Curiosity



(a) Distant (Average-type Minimizer)



(b) Close (Intersection-type Minimizer)

Figure from [Nexus Optimizer](#)

observe

explain



predict

control

➤ Vision

What we can do to the system that would make the principle reveal whether it is true?
If we understand a dynamical mechanism well enough, we should eventually be able to control it.

01

Mechanics
of Learning

What governs the trajectory?

02

Scaling
of Learning

Which principle is universal?

03

Dynamics
of Evolution

How does new learning interact
with existing knowledge?

04

System
Support

How should learning be
engineered across the computing?

Thanks :)